Teaching Statement

Showing students the work behind every answer, then asking for theirs

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In my first semester as a teaching assistant, a student wrote on my evaluation that asking me questions was "like talking to a chatbot." The student granted that I knew a lot, but nobody could follow me. I had been giving answers and keeping the work behind them in my head. Since then my teaching has been about showing the work. Students see how I write code and approach a problem, they do the same work themselves, and then they show me theirs.

Feedback taught me to write out my reasoning

The chatbot comment came from first-year PhD Macroeconomics I in Fall 2022, where students rated my clarity at 2.67 out of 5 and my pace at 2.33. They also found my handwritten notes hard to read. I started pitching explanations to where students were, working through homework with them in office hours, and typing full step-by-step solutions to each problem set, which students could redo line by line the way they would rerun a notebook. In Macroeconomics II the next semester, both ratings rose to between 4.38 and 4.50 across the two sections, and one student singled out my "clear explanation and detailed solution." I then carried the same approach into code as a teaching assistant for two graduate courses outside my field, Advanced Microeconometrics and Structural Microeconometrics. I led the sections in Python, wrote the solution code for the problem sets and graded students' code. When the course reached dynamic discrete choice models, I walked the class through the estimation code one step at a time.

Students code along instead of watching

In Spring 2025 I designed and taught my own courses: Econ 4850, Economic Analysis with Python, for upper-level undergraduates, and Econ 6850, A First Python Course for Economists, for graduate students. Both were half-semester introductions for students with little or no programming experience. I ran them as flipped classrooms in which students followed along on their own laptops, and each class meeting included short exercises on the code just introduced. I walked through each exercise with the class and took questions as they came, so a student writing a first loop got the same practice as one who had programmed before. What I enjoy most in a flipped classroom is sharing my own coding practice and the way I approach a problem. Feedback still shapes these courses. In Econ 6850, only two of the nine enrolled students responded to the course evaluation. Both rated every item on learning, organization, enthusiasm and rapport at 5 out of 5, but one called the course "not for beginners," although it is meant for beginners. Next time, students will complete a short diagnostic exercise before the first class, and those without programming background will work through a setup module first.

AI comes in once students can read its code

An AI assistant hands students finished code without the work behind it, so I staged when they could use it. In the first weeks, while students learned syntax, I asked them to turn off AI autocompletion. Once we moved on to quantitative exercises, they turned it back on to work faster, and in the final weeks I asked them to use AI as much as they could. Students will use AI whatever the syllabus says. A ban would have trained a skill that AI is likely to displace and would have been costly to enforce, while open use from the first day would have let students hand in code they could not read. I also built the written assignments to be incentive compatible. Part of each grade rewarded

completion, and I broke problems into small steps, which made it cheaper for students to work a problem through themselves than to outsource it.

Students show their work and defend it

Each course ended with a short group research project. I met with each group to settle the project's question and method. Before presenting, the group sent me its slides and Jupyter notebook, and I read both to prepare questions about the group's own code and output. One group of undergraduates in Econ 4850 compared a hedonic pricing model of more than 44,000 Canadian housing listings with a neural network. Before fitting the network, the students computed the OLS coefficients with matrix algebra and checked them against statsmodels, then estimated them again by gradient descent, which avoids inverting a large matrix and is how the network itself is trained. They then asked what the network gained in predictive accuracy and lost in interpretability.

An elective built on Python labs

I am prepared to teach intermediate macroeconomics, the PhD macroeconomics sequence, econometrics and computational economics. I would also like to teach a new undergraduate elective, Inequality, Entrepreneurship and the Macroeconomy, that draws on my research on how entrepreneurs and venture-capital financing shape the top of the wealth distribution. The course would require intermediate macroeconomics and introductory statistics, and it would pair each of four units with a Python lab. Students would measure top wealth shares in the Survey of Consumer Finances, estimate the Pareto tail of the wealth distribution, describe venture-capital deal data, and simulate a two-type heterogeneous-agent model to evaluate a tax on top wealth. Readings would combine recent research articles with three online texts: the QuantEcon lectures by Thomas J. Sargent and John Stachurski, Simon Scheidegger's *Deep Learning for Solving and Estimating Dynamic Models in Economics and Finance*, and *Applied Causal Inference Powered by ML and AI* by Chernozhukov, Hansen, Kallus, Spindler and Syrgkanis. Students would hand in each lab as a notebook and finish with a group research project that they present and defend, as in my Python courses.

Teaching Experience, Washington University in St. Louis

Role	Course	Level	Term
Instructor	Econ 4850 Economic Analysis with Python	Undergraduate	Spring 2025
Instructor	Econ 6850 A First Python Course for Economists	Graduate	Spring 2025
Teaching assistant	Econ 501 Macroeconomics I	PhD	Fall 2022
Teaching assistant	Econ 502 Macroeconomics II (two sections)	PhD	Spring 2023
Teaching assistant	Econ 5141 Advanced Microeconometrics	Graduate	Spring 2024, Fall 2024
Teaching assistant	Econ 5175 Structural Microeconometrics	Graduate	Spring 2024